Ch 7/8 Potential Quiz and Module 2 Exam Answers

A subset of these questions will appear on the Chapter 7/8 Quiz.

Quizzes are 30 minutes long and have 20 points.

A (probably different) subset of these questions will also appear on the Module 2 Exam, with questions from the Ch 6 and Ch 7/8 Quiz lists together making up at least half the points on the Module 2 Exam (i.e. at least 25 out of 50 points)

You should not expect Quiz or Exam questions to appear in the same order as in this document, nor should you expect the options for the answers to appear in the same order as in this document.

The remining 50% of Module 2 Exam questions will be rephrasing or recombination of topics listed either in this document, in the Chapter 6 Potential Quiz questions, or the Ask Your Neighbor questions from class.

Questions are organized below into "pools" of similar questions; for each question it is shown how many points each of those questions would be worth should the question appear on the exam.

Answers are included for non-numeric questions in this document, but you should not count on being able to memorize an answer key for the exam, and you will not be allowed access to notes during the exam. You are strongly encouraged to form study groups to prepare for the exam, and you are encouraged to ask Gwen if you don't understand any concepts that are in this study sheet.

There is only one pool for this quiz

Pool 1: Each worth 1 point

1. Suppose a poor economy inches towards the steady state in Solow's exogenous growth model. What happens?

A. The growth rate of output decreases.
B. The depreciation rate increases.
C. Saving per capita decreases.
D. Consumption per capita decreases.

Answer: A

2. Use the endogenous human capital accumulation model from Chapter 8 to consider the following real-world situation:

Imagine that internet access improves for people living in remote and under-developed regions of the world; Less-Developed Economy Consumption would likely _____ in the current period and be on _____ growth path as a result.

A. be unchanged; steeper
B. fall; steeper
C. be unchanged; the same
D. fall; the same

Answer: A

3. According to the assumptions of the Chapter 8 endogenous growth model, would an educated worker prefer to live in a country with other highly-educated people or in a country where most other people are not highly educated?

A. Prefer to live in the country with high education levels in the short run but not the long run.
B. Prefer to live in the country with low education levels in both the short run and long run.
C. Prefer to live in the country with high education levels in both the short run and long run.
D. Prefer to live in the country with low education levels in the short run but not in the long run.

Answer: C

4. In the Golden Rule steady state, the marginal product of capital is equal to the

A. savings rate plus the population growth rate.
B. population growth rate plus the savings rate.
C. population growth rate plus the depreciation rate.
D. savings rate plus the depreciation rate.

Answer: C

5. Which, if any, of the following statements is not correct?
A. Per-capita standards of living were fairly constant everywhere in the world until the early 1600s.
B. Per-capita standards of living started to rise most dramatically starting with the Industrial Revolution.
C. Per-capita standards of living have increased most everywhere in the world since the 1960s.
D. All of these statements are supported by the data.
Answer: D

6. In the Malthusian model, an improvement in the technology of growing food is most likely to
A. increase the equilibrium size of the population and increase the equilibrium level of consumption per worker.
B. increase the equilibrium size of the population and decrease the equilibrium level of consumption per worker.
C. increase the equilibrium size of the population and have no effect on the equilibrium level of consumption per worker.
D. have no effect on the equilibrium size of the population and increase the equilibrium level of consumption per worker.
Answer: C

7. The endogenous growth model from Chapter 8 makes the assumption that education is non-rival. That implies that
A. knowledge is not in limited supply.
B. there is more than one way to produce the same good.
C. the market for education is not competitive.
D. there are externalities associated with an individual's education decision.
Answer: A

8. Predictions of the Malthusian growth model were wrong because:
A. The industrial revolution added capital to the production function which is not in fixed supply.
B. The model didn’t consider technological progress in the production function.
C. The model didn’t consider population growth in the production function.
D. Having children is never an economic decision.
Answer: A

9. Which of the following statements is true about the Solow model in the long run?
A. The standards of living of a country depend on the savings rate but not on the population growth.
B. The standards of living of a country depend on population growth but not on the savings rate.
C. The standards of living of a country depend on the savings rate and population growth.
D. The standards of living of a country depend on neither population growth nor the savings rate.
Answer: C

10. The Solow residual attempts to measure the amount of output not explained by
A. technological progress.
B. the direct contribution of labor and capital.
C. economic forecasts.
D. a nation's level of education.
Answer: B

11. Suppose that government spending makes human capital technology more efficient. Use the graphical model from Chapter 8 to show how a lump-sum tax on Consumption to improve education would affect new consumption path. Discuss the conditions under which this policy might be optimal from a social planner's perspective and why it might not be chosen by the market even if it is more efficient.
In this case we expect
A. The graph of ln(Ct) to shift up and get steeper.
B. The graph of ln(Ct) to get steeper but not shift.
C. The graph of ln(Ct) to get flatter but not shift.
D. The graph of ln(Ct) to shift down and get steeper.
Answer: D

12. Which of the following is NOT positively correlated with per capita GDP growth?
A. Widening of the income distribution.
B. All of these are positively correlated with per capita GDP growth.
C. Literacy rates, adult.
D. Average number of years in education, female.
Answer: A

13. If the savings rate falls in the Solow growth model,
A. steady state capital per worker rises.
B. output growth is unchanged in the steady state.
C. per worker output is unchanged in the steady state.
D. per capita consumption is unchanged in the steady state.
Answer: B

14. Statistically, countries that have _____ barriers to international trade and international finance have _____ gains in per capita GDP and _____ average life expectancies.
A. greater, greater, longer
B. greater, lower, shorter
C. lower, greater, shorter
D. lower, lower, longer
Answer: B

15. The idea that an improvement in technology causes an increase in population but causes no increase in the average standard of living is attributed to
A. Adam Smith.
B. Thomas Malthus.
C. Robert Solow.
D. Paul Romer.
Answer: B

16. Recent evidence suggests that output per worker is
A. positively related to both the rate of investment and to the rate of population growth.
B. positively related to the rate of investment and negatively related to the rate of population growth.
C. negatively related to the rate of investment and positively related to the rate of population growth.
D. negatively related to both the rate of investment and to the rate of population growth.
Answer: B

17. A country with low literacy rates institutes a law that says people need to go to school and pass a literacy test before being able to work. Using the endogenous growth model this will:
A. Shift the consumption path down.
B. Shift the consumption path up.
C. Shift the consumption path down, but pivot it upwards.
D. Shift the consumption path up, but pivot it downwards.
Answer: C

18. A steady state is
A. a temporary equilibrium.
B. a Pareto Optimum.
C. a long-run equilibrium.
D. an economy with ongoing fluctuations.
Answer: C

19. In the Malthusian model, state-mandated population control policies are likely to
A. decrease the equilibrium size of the population and increase the equilibrium level of consumption per worker.
B. decrease the equilibrium size of the population and have no effect on the equilibrium level of consumption per worker.
C. have no effect on the equilibrium size of the population and increase the equilibrium level of consumption per worker.
D. have no effect on either the equilibrium size of the population or the equilibrium level of consumption per worker.
Answer: A

20. In the Malthusian model, improvements in health care lead to
A. higher population and higher per-capita production.
B. higher population and lower per-capita production.
C. lower population and higher per-capita production.
D. lower population and lower per-capita production.
Answer: B

21. In more modern times as opposed to the times of Malthus, higher standards of living appear to
A. decrease death rates and increase birth rates.
B. decrease death rates and also decrease birth rates.
C. decrease death rates and have no effect on birth rates.
D. have had effects on neither death rates nor birth rates.
Answer: B

22. All of the following increase total factor productivity except
A. new inventions.
B. more capital.
C. new management techniques.
D. favorable changes in government regulations.
Answer: B

23. The saving rate has the following characteristic in Solow's exogenous growth model
A. It increases with output.
B. It first decreases, then increases with output.
C. It first increases, then decreases with output.
D. It is constant.
Answer: D

24. In the Solow growth model, long run growth in the standard of living is propelled by
A. government spending.
B. technological change.
C. growth in the capital stock.
D. growth in the labor force.
Answer: B

25. In Solow's exogenous growth model, the economy reaches a stable steady state because
A. the marginal return of capital is decreasing.
B. capital is growing at a constant rate.
C. the substitution effect is stronger than the income effect.
D. conditional convergence holds.
Answer: A

26. In the steady state of Solow's exogenous growth model, an increase in the savings rate
A. increases output per worker and increases capital per worker.
B. increases output per worker and decreases capital per worker.
C. decreases output per worker and increases capital per worker.
D. decreases output per worker and decreases capital per worker.
Answer: A

27. In the steady state of the Solow model and using the notation from class, per capita GDP (y) is
A. increasing.
B. constant.
C. decreasing.
D. we do not know without knowing exogenous variables such as z, n, and depreciation rate (d).
Answer: B

28. In Solow's exogenous growth model, the steady-state growth rate of capital can be increased by
A. higher population growth.
B. higher depreciation rate.
C. higher saving rate.
D. higher interest rate.
Answer: A

29. The Golden Rule of capital accumulation maximizes the steady-state level of
A. output per worker.
B. capital per worker.
C. consumption per worker.
D. investment per worker.
Answer: C

30. In the steady state of Solow's exogenous growth model, an increase in total factor productivity
A. increases output per worker and increases capital per worker.
B. increases output per worker and decreases capital per worker.
C. decreases output per worker and increases capital per worker.
D. decreases output per worker and decreases capital per worker.
Answer: A

31. The Solow growth model can account for
A. the patterns of international trade.
B. business cycles.
C. why richer countries have higher investment rates.
D. the role of government in spurring growth.
Answer: C

32. Income per worker has been
A. converging in both the rich countries and the poor countries.
B. converging in the rich countries, but not converging in the poor countries.
C. converging in the poor countries, but not converging in the rich countries.
D. converging in neither the poor nor the rich countries.
Answer: B

33. All other things being equal, if the population growth rate is lower, then the golden rule level of the per capita capital stock will be
A. higher.
B. lower.
C. unchanged.
D. indeterminant (either higher, lower, or unchanged).
Answer: A

34. Which of the following is not a reason for differences in total factor productivity across countries?
A. Differences in the size of population.
B. Learning by doing.
C. Barriers to the adoption of technology.
D. Inefficient allocation of factors of production across firms in some countries.
Answer: A

35. In the endogenous growth model, a decline in total factor productivity (z) is represented by
A. a downward shift in the Consumption growth path.
B. a movement to the left along the Consumption growth path.
C. a flatter Consumption growth path.
D. all of these things.
Answer: A

36. In the endogenous growth model, workers divide their time between market work and
A. accumulating physical capital.
B. accumulating human capital.
C. trying to invent production processes.
D. work at home.
Answer: B

37. Endogenous growth theory is about
A. the welfare of indigenous people.
B. explaining the importance of history on economic growth.
C. studying how fertility choices impact standards of living.
D. giving more importance to capital accumulation.
Answer: B

38. In the endogenous growth model, education is best described as a type of
A. all of these things.
B. investment.
C. government spending.
D. production.
Answer: B

39. What characteristic do both human and physical capital share?
A. Both are controlled by the government.
B. Current costs are incurred for future benefits.
C. The use of both exhibits rivalry.
D. They can easily be transferred to ownership.
Answer: B

40. Human capital is knowledge in
A. books.
B. people.
C. firms.
D. government.
Answer: B

41. In the development of the endogenous growth model from class, Paul Romer argued that a key feature of knowledge is
A. divisibility.
B. private ownership.
C. nonrivalry.
D. durability.

Answer: C

42. A key characteristic of the production function in the endogenous growth model presented in the text is that
A. there are increasing returns to scale.
B. there are decreasing returns to scale.
C. there are constant returns to scale.
D. at low levels of human capital, there are increasing returns to scale in human capital, while at high levels of human capital, there are decreasing returns to scale in human capital.

Answer: C

43. The endogenous growth model predicts that
A. there is convergence in incomes per capita across countries.
B. output per capita is constant.
C. rich countries will always become poor.
D. differences in per capital incomes across countries persist forever.

Answer: D

44. The BRICS countries are
A. Brazil, Romania, India, Colombia, and Surinam.
B. Belarus, Romania, Indonesia, China, and Singapore.
C. Brazil, Russia, India, China, and South Africa.
D. Britain, Russia, Iceland, Cypress, and Spain.

Answer: C

45. In the endogenous growth model,
A. households can choose their own level of income.
B. the growth rate of output increases with the savings rate.
C. individual household choices affect the incomes of other households around them.
D. all of these things.

Answer: C

46. Globalization is thought to contribute to higher per capita GDP because
A. of the efficiency gains associated with specialization.
B. of the efficiency gains associated with more competition.
C. of the efficiency gains associated with access to foreign savings and investment opportunities.
D. all of these things.
Answer: D

47. According to the data from class, the economic costs associated with US droughts and storms have been rising over the past 40 years. Assuming we are at the golden level of per capita consumption, increased frequency of these types of climate events would be most consistent with ___ levels of per-capita savings.
A. lower
B. higher
C. no change in the
D. ambiguous changes in the
Answer: A

48. If there are human capital externalities, then
A. human capital should be taxed.
B. convergence in per capita incomes occurs.
C. differences in human capital across countries can persist.
D. pollution is a problem.
Answer: C

49. In the endogenous growth model presented in the text,
A. consumption grows faster than human capital.
B. human capital grows faster than consumption.
C. both consumption and human capital grow at the same rate.
D. neither consumption nor human capital grows in the steady state.
Answer: C

50. An example of an increase in b, the efficiency of human capital accumulation, is
A. more mandatory school years.
B. more population growth.
C. better school material.
D. better total factor productivity.
Answer: C

51. Schooling is subsidized in the United States in part
A. so that the golden rule capital level can be reached.
B. because otherwise people would not get educated.
C. because there is a constitutional right to education.
D. because there is a positive externality from schooling.
Answer: D

52. An improvement in school quality translates into an increase in which model parameter?
A. H
B. z
C. b
D. 1-u
Answer: C

53. What immediate consequence does an increase in education time have in the endogenous growth model with human capital?
A. Lower output.
B. Lower output in the future.
C. Lower wages.
D. Lower human capital.
Answer: A

54. In the Malthus growth model, an increase in population growth results in a/an _____ in per capita GDP, whereas in the Solow growth model, an increase in population growth results in a/an ____ in per capita GDP.
A. increase; increase
B. increase; decrease
C. decrease; increase
D. decrease; decrease
Answer: D

55. Assume the following capital accumulation equation from class: K'=(1-d)*K + I.
Starting from a steady-state equilibrium, assume that Investment spending permanently increases by amount "x". The total change in the capital stock K will be approximately
A. d*x.
B. (1-d)*x.
C. x/d.
D. x/(1-d).
Answer: C